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NSF Protocol P248 Military Operations Microbiological Water Purifiers

February 2012

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NSF Protocol P248
Military Operations
Microbiological Water Purifiers

Protocol Developer

Prepared by the U.S. Army Public Health Command [USAPHC, formerly the U.S. Army Center for Health Promotion and Preventive Medicine (USACHPPM)] using a grant received from the Army Study Program Management Office.

Published by:
NSF International
789 Dixboro Road
Ann Arbor, Michigan 48105

Printed in the United States of America.

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U.S. ARMY PUBLIC HEALTH COMMAND
ARMY INSTITUTE OF PUBLIC HEALTH
PORTFOLIO, ENVIRONMENTAL HEALTH ENGINEERING
WATER SUPPLY MANAGEMENT PROGRAM

AND

NSF INTERNATIONAL

PROTOCOL FOR EVALUATING THE MICROBIOLOGICAL TREATMENT CAPABILITIES OF SMALL WATER PURIFIERS INTENDED FOR USE DURING MILITARY OPERATIONS

February 2012

PREFACE

In the early 2000s, scientists and engineers in the Department of Defense (DOD) identified a mission critical need to assess the performance of commercial off-the-shelf (COTS) individual water purification devices. The military-issued emergency water purifiers in use since the World War II era required multiple time-consuming steps, and could in some cases provide a false sense of security, actually failing to produce potable (microbiologically safe for human consumption) water from available water sources, with no indication that such was the case. The comparatively recent appearance of a multitude of hand-held COTS individual water purifiers (IWPs) was perceived by many to be the remedy to the problem. According to the commercial advertising campaigns, these devices were able to produce microbiologically safe and palatable water from nearly any quality water source, and could keep Warfighters hydrated and mission-ready when they did not have access to military-provided bulk water supplies. COTS IWPs began to proliferate among deploying units that were able to purchase them with unit charge cards, and among individual deploying Servicemembers who would obtain them at their own expense.

None of the IWPs had undergone rigorous DOD evaluations of their effectiveness and applicability to military missions, nor had they been evaluated and approved by any of the Services' Surgeon Generals. The primary concern for the lack of military and medical endorsement was for microbiological contaminants – bacteria, viruses, and protozoan cysts – that could rapidly reduce a Warfighter's mission readiness through acute gastro-intestinal distress or worse. The concern presented an unacceptable risk to deployed personnel in the eyes of the members of the Joint Medical Field Water Subgroup, a subgroup to the Joint Environmental Surveillance Workgroup, which was chartered under DOD Health Affairs. Consequently, a project was initiated to develop a test protocol that all IWPs intended to be marketed to the DOD could be subjected to, to determine their effectiveness in providing microbiological purification. Devices that successfully completed the test protocol could then be used by deployed individuals with a greater measure of assurance that the water they would obtain by using the devices would be of acceptable microbiological quality and would not cause acute illness or disease.

Many current military operations lack the centralized nature of conventional warfare. The transformation to a more agile fighting force where Warfighters inhabit tactical, unit-sized Forward Operating Bases (FOBs), smaller Command Outposts (COPs), and safe-houses, creates unique, never-before-encountered requirements for field water operations and surveillance. These distributed operations have created a gap in fielded materiel for treating water at the unit level (10-50 personnel). Similar to IWP procurement, deploying units have looked to the commercial market to purchase treatment systems [termed small unit water purifiers (SUWPs)] sized between individual purifiers and the U.S. military-fielded Lightweight Water Purifier (LWP), incorporating reverse osmosis and capable of producing 125 gallons per minute from fresh water sources.

This Protocol has been developed with the firm hope that vendors who desire to market their individual or small unit water purification devices to the military will use it to develop, evaluate, and improve their devices. Additionally, it is recommended for use during government down-selection and combat development efforts to determine device capabilities. The ultimate goal is to provide our Warfighters the capability to produce sustainable quantities of microbiological contaminant-free drinking water, when necessary, from any fresh-water source that may be available to them during deployments.

The U.S. Army Public Health Command (USAPHC) received a grant from the Army Study Program Management Office to develop this Protocol during FY 2005. USAPHC coordinated closely with personnel from NSF International who provided much guidance and assistance in developing the Protocol and who graciously offered to publish it. USAPHC also solicited and received input from many other DOD organizations, who contributed significantly to the refinement of this Protocol, including the U.S. Navy Environmental Health Center, the U.S. Navy Bureau of Medicine, the U.S. Air Force Institute for Operational Health, the U.S. Air Force 311th Human Systems Wing, the U.S. Army Proponency Office for Preventive Medicine, the U.S. Army Deputy Chief of Staff G-4, DALO-SMT, and the U.S. Army Infantry Center and School. The many efforts of individuals from all of these and other organizations are greatly appreciated
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SECTION 1
INTRODUCTION

1.1 BACKGROUND

1.1.1 The need to purify natural waters of unknown or variable microbiological quality is common among members of the military and non-military government agencies, consumer groups, manufacturers, campers, and hikers, as well as others who do not always have ready access to a treated water supply. In response to this need, entrepreneurs and manufacturers have developed a number of small water purifiers [SWPs, to include individual water purifiers (IWPs) and small unit water purifiers (SUWPs)] for campers, hikers and non-government organizations conducting humanitarian assistance operations. Insofar as these devices are intended for military field use, their performance must be thoroughly evaluated to allow informed product selection, and to protect the health of the Warfighters that use them.

1.1.2 This Protocol was derived and adapted primarily from existing publications of the U.S. Environmental Protection Agency (USEPA) and NSF International (references 1, 2 and 3). It describes the procedures to be used to test SWPs designed to achieve reduction or inactivation of microbiological contaminants, including bacteria, viruses, protozoa, and protozoan cysts and oocysts from virtually any fresh water source likely to be encountered by military personnel during exercises and deployments.

1.1.3 Those testing to this Protocol may seek two distinct endorsements. Products may be "certified" by NSF International as having met the requirements of this Protocol and may be granted privilege to use the NSF "mark" when marketing their product. All requirements of this protocol must be met. Contact NSF International for more information. Similarly, execution of this Protocol for acceptance of an SWP for military use will be administered and monitored by a Government Review Agency (GRA) designated by the Department of the Army Office of the Surgeon General. The GRA may grant "compliance" with Appendix B of this Protocol. Those seeking NSF "certification" and GRA "compliance" should convey this request prior to test initiation.

1.1.4 For the purposes of this Protocol, and with reference to their use by deployed military personnel, SWPs for microbiological purification are not intended to significantly reduce chemical contaminants [toxic industrial chemicals (TICs), biological or chemical warfare agents]. Claims made by vendors concerning chemical contaminant reduction efficiencies are not tested and will not be verified by following this Protocol. IWPs tested using this Protocol are intended for individual use during emergency or short term planned missions as an alternative to the currently fielded Globaline® iodine tablets and Chlor-Floc® disinfection supplies. SUWPs tested against this protocol may be used during emergency and short term planned missions (less than 7 days), and potentially during long term operations provided produced water quality and field medical oversight follows current doctrine.

® Globaline is a registered trademark of Wisconsin Pharmacal Company, Jackson, WI.
® Chlor-Floc is a registered trademark of Deatrick & Associates, Inc., Alexandria, VA.

1.1.5 A purifier specific test plan (PSTP) will be developed for each purifier or type of purifier based on this Protocol. The Protocol is intentionally general in nature so as to be applicable to all SWPs, irrespective of physical construction or technology employed. If users find procedures described in this Protocol inappropriate or inapplicable to adequately test their particular purifier or technology, they should describe proposed deviations and document the reasons for modifying or departing from the Protocol in the PSTP. Such deviations and departures will be evaluated by the GRA and judged acceptable as long as the level of testing is not decreased, nor the intent of the Protocol altered. Test conditions and procedures should be designed to specifically challenge the particular technology under evaluation. The testing of SWP performance involves five sequential steps:

(1) development, submission to GRA, and approval of the PSTP
(2) execution of third-party testing in accordance with (IAW) the PSTP
(3) data reduction and analysis
(4) report preparation and information transfer
(5) performance review by GRA and issuance of a Letter of Compliance

1.1.6 Test Sponsors (to include vendors/manufacturers/non-government and government organizations) who's devices successfully comply with the performance criteria of this Protocol, meet Appendix B requirements, and submit the results to the GRA will receive a GRA P248 Letter of Compliance stating that the device meets the military requirements of NSF Protocol P248, *Military Operations Microbiological Water Purifiers*. Without the Letter of Compliance test sponsors shall not claim, or market their device as being compliant with Appendix B of this protocol. Information on requirements for GRA compliance can be found in Appendix B.

1.1.7 Application of SWPs intended for individual and group use in non-military operations is within the scope of this Protocol. Organizations may request assistance from the GRA for non-military applications.

1.1.8 Inasmuch as this Protocol was developed subsequent to and is based heavily on the USEPA Guide Standard (reference 1) and NSF Protocol P231 (reference 3), and some vendors of SWPs have already tested their purifiers, either in their own laboratories or by a third-party, using these protocols, those vendors may submit a request to the GRA that a device be "grandfathered" into compliance. On receipt of such a request, together with a copy of the test plan followed and the evaluation and report or results, the GRA will review the documents and determine the equivalency of the tests performed to those required by this Protocol, and at its sole discretion, grant or deny grandfathering. If the GRA grants grandfathering, it will provide the vendor a letter stating that they may claim NSF Protocol P248 compliance for their device(s). In the case of denial, the GRA will issue a written statement to the vendor specifying the reasons for denial, and, in any case, the vendor may retest the purifier(s) using this Protocol and/or appeal the denial with rebuttal and/or additional information for the GRA to reconsider grandfathering as a means of compliance.

1.1.9 To promote consistent, accurate, and comparable results across laboratories, all laboratories performing testing of water treatment equipment for U.S. Military application under

NSF Protocol P248 are required to possess / obtain accreditation based on the International Organization for Standardization / International Electrotechnical Commission (ISO/IEC) 17025:2005 (reference 4), by a recognized third party accrediting organization [e.g., American Association for Laboratory Accreditation (A2LA), National Environmental Laboratory Accreditation Program (NELAP), etc.]. All analytical procedures conducted during protocol execution for reporting must be listed on the laboratory's "Scope of Accreditation". These requirements also apply to contract laboratories conducting analyses on behalf of the primary testing laboratory. ISO/IEC 17025 is an internationally-accepted standard for developing laboratory management systems for quality, administrative, and technical operations and an accreditation is a crucial requirement for acceptance of test data by the U.S. Military.

1.2 EQUIPMENT TESTING RESPONSIBILITIES

1.2.1 Test Sponsor Responsibilities. The test sponsor is responsible for:

- notifying the GRA of the intent to test equipment using this Protocol
- selecting, negotiating with, and obtaining the concurrence of the GRA for the third-party equipment testing organization (ETO)
- developing the PSTP (with assistance from the ETO)
- submitting the PSTP to the GRA for review and comment
- providing new, commercially-packaged purifiers to the ETO for testing, complete with instructions
- providing all necessary resources, including monetary, logistical, and technical support to the ETO to enable testing to proceed on schedule
- ensuring that the final report of testing, in its entirety, is forwarded to the GRA (it is expected that test results will be provided to the GRA regardless of device performance so that the GRA can inform potential users of the health risks of using the device.)

1.2.2 ETO Responsibilities. The ETO is responsible for:

- Providing documentation on accreditation to ISO/IEC 17025 in accordance with section 1.1.8 of this protocol.
- assisting the test sponsor in developing the PSTP and having it approved by the GRA
- coordinating with the GRA to resolve product- and technology-driven modifications to the Protocol
- conducting the testing IAW the GRA-reviewed and approved PSTP
- managing, documenting, evaluating, interpreting, and reporting all test results
- providing a written report documenting the test procedures, deviations from this Protocol, results of testing, and the performance evaluation of the IWP as directed in the PSTP.

- providing quality assurance and documentation as requested by the GRA

1.2.3 GRA Responsibilities. The GRA is responsible for:

- assisting test sponsors and ETOs in interpreting and complying with this Protocol
- assisting test sponsors, if necessary, in identifying and selecting qualified ETOs
- reviewing and approving PSTPs, in a timely manner (a submitted PSTP that remains unreviewed for a period greater than 15 working days after receipt by the GRA will be considered approved by default)
- reviewing the laboratory test results and providing a Letter of Compliance to the test sponsor explaining the basis for acceptance or rejection of the equipment tested
- coordinating with the appropriate military offices/officials to publish the results of testing throughout the Services and DOD, as required

1.3 PSTP DEVELOPMENT, REVIEW, AND APPROVAL

1.3.1 The ETO and the test sponsor must adhere to the requirements of this Protocol in developing the PSTP. The final draft PSTP submitted to the GRA for review and approval shall:

- include the information requested in this Protocol
- conform to the format identified herein
- conform to the testing procedures described herein
- include any unique requirements identified by the GRA during initial coordination

1.3.2 Specifically, the PSTP shall include the following essential elements (see Appendix C for a PSTP template):

- the test sponsor's name and address, and a point-of-contact (POC) name and telephone number
- the name and address of the ETO and a list of designations, recognitions, certifications, etc., that qualify the ETO to perform the testing
- the names, model numbers, and serial numbers of the tested IWPs
- the manufacturer's operating instructions (as an attachment)
- a description of the roles and responsibilities of the testing participants
- a description of the SWP, the basic technologies it uses, and its advertised capabilities
- A list of materials for all parts that contact water, to include acceptance of materials for contact with drinking water and/or sufficiently detailed material specifications so that a toxicological analysis can be performed.

- the experimental design for the actual testing of the SWP, including any variations from the Protocol requirements
- a description of the procedures governing testing activities such as equipment operation and process monitoring; sample collection, preservation, and analysis; and data collection and interpretation

1.3.3 Once the ETO and the test sponsor have an agreed-upon draft PSTP, it shall be submitted to the GRA for technical review and approval prior to initiation of any testing.

SECTION 2
PERFORMANCE REQUIREMENTS

2.1 MICROBIOLOGICAL WATER PURIFICATION

A SWP must demonstrate the microbiological reduction capabilities described in Table 2-1 when tested according to an approved PSTP developed IAW this Protocol as described in Section 3.

Table 2-1 – Microbiological Reduction Requirements.

Organism	**Influent Challenge**[1]	**Minimum Required Reduction**[2]	
		Log	**%**
Bacteria: *Escherichia coli*, or *Raoultella terrigena*, or *Bacillus atrophaeus* (spore form)	$\geq 10^7$/100 mL	6	99.9999
Virus: MS2 and fr coliphages	$\geq 10^7$/L	4	99.99
Cyst: *Cryptosporidium parvum* oocysts	$\geq 5x10^4$/L	3	99.9

[1] The influent challenges may constitute greater concentrations than would be anticipated in source waters, but these are necessary to properly test, analyze, and quantitatively determine the indicated log reductions. The influent challenge must not be less than that required to demonstrate the minimum required log reduction. Mathematical rounding is not allowed in determination of overall log reduction.
[2] "Reduction" is kill, remove, or inactivate.

2.1.1 Devices that incorporate ultra-violet (UV) light, without additional treatment technologies for pathogen reduction, shall use MS2 bacteriophage at an influent concentration of 5×10^4 to 5×10^5 per L. Testing shall follow NSF/American National Standards Institute (ANSI) Standard 55 collimated beam procedures to measure virus stock susceptibility to UV light and determine the resultant log reduction of MS2 that equates to a dose of 40 mJ/cm^2. This log reduction then becomes the reduction requirement for Protocol compliance. UV testing shall incorporate all other applicable requirements of this Protocol. UV devices that do not include a vessel for water during treatment must show pathogen reduction efficacy using the most current and appropriate vessel available to military personnel (e.g., canteen or personal hydration system). Compliance with Protocol requirements, if shown, must state the water vessel used during testing.

2.1.2 Devices that incorporate un-common, or multiple technologies for pathogen reduction/inactivation may require different organisms and/or procedures than described in this Protocol to fully demonstrate device capabilities. Unless it has been shown that one of the listed non-pathogenic surrogates is an acceptable challenge for the given technology, the use of the naturally occurring pathogen is required. The PSTP will provide a forum for the manufacturer to provide specific testing procedures for these technologies to the GRA for approval. NSF Protocol P231 (reference 3) and the USEPA Guide Standard (reference 1) may be consulted for the selection of appropriate organisms and/or procedures to conduct testing.

2.2 OPERATIONAL CAPABILITY

2.2.1 While this Protocol does not presume to dictate the physical characteristics of SWPs that make them useful to or desirable by Warfighters – those characteristics are defined by units, unit commanders, and missions – there are desirable characteristics and operational capabilities that will be considered in the final recommendation as to device suitability for use in various military field operation scenarios.

2.2.2 The characteristics listed below have been selected from current and draft operational requirements documents (reference 5), and are considered desirable, if not required by the U.S. Military. Commanders planning to purchase specific SWPs will consider these or similar criteria and how they relate to the mission for which the SWPs are to be selected in their decisions. Not all of the following are evaluated by this Protocol. The ideal SWP would, in no particular order of importance:

- allow filling and servicing without contaminating the purified water or surfaces that contact the purified water
- be able to be worn with, attached to, or carried on or in current and future load bearing equipment (IWP only)
- be capable of operation by Warfighters in any terrain type and climatic condition (Basic and Hot)
- be fully operational after freeze / thaw cycles
- be compatible with standard 1- and 2-quart canteens and military-approved personal hydration systems
- be easily cleaned and disinfected
- be operable by military personnel of all sizes, strengths, and physical condition
- have a functional life after initial use of at least 120 days (IWP only), and a shelf life of 5 years
- have a treat-to-drink time of less than 20 minutes
- provide an explicit indication or assurance of the unit's effective use lifetime (in terms of volume purified) to warn the consumer of potential diminished treatment capability through one or more of the following:
    - cessation of water flow
    - a visual change
    - simple, explicit instructions for servicing and replacing components within the recommended use life (measurable in terms of volume throughput, specific time frame, or other appropriate method)
- purify at least 135 liters of water as packaged (IWP only)
- reduce water turbidity to less than 1 nephelometric turbidity unit (NTU)

- require little or no training to operate and maintain
- resist mold, fungus, and biofilm growth
- use readily available and easily installed replacement filters and parts

2.3 BIOCIDAL CHEMICALS

2.3.1 Biocidal Chemical Leaching

Where silver or other biocidal chemicals are used in a candidate SWP, the concentrations of those chemicals in the product water must be less than: 1) the corresponding short-term Military Exposure Guidelines (MEG) concentrations in the current edition of USAPHC Technical Guide (TG) 230 (reference 6); 2) the corresponding USEPA National Primary Drinking Water Regulation (NPDWR) maximum contaminant levels (MCL) if there are no MEGs, or otherwise be demonstrated not to constitute a threat to health from consumption or contact if neither MEGs nor MCLs exist.

2.3.2 Stability of Biocidal Chemical

Where biocidal chemicals are used in the treatment unit, the stability of the chemical for disinfectant effectiveness should be sufficient for the potential shelf life and the projected-use life of the unit based on manufacturer’s data. Where stability cannot be assured from historical data and information, additional tests may be required.

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SECTION 3
MICROBIOLOGICAL WATER PURIFIER TEST PROCEDURES

3.1 PURPOSE

The candidate SWP is tested to substantiate its microbiological reduction or inactivation capabilities as defined in Table 2-1 over the effective use life of the device and, where a biocidal chemical is used, to determine that said chemical is not present in the effluent at excessive levels.

3.2 APPARATUS

Three production units of each candidate SWP will be tested simultaneously. The units should be new and removed from the commercial packaging just prior to beginning testing or, if appropriate, shall be "reset" with new, production components to perform as new systems in the existing commercial configuration. Manufacturers are welcome to test research and development (R&D) or "Beta" devices against this Protocol, however, GRA involvement, including PSTP and subsequent laboratory data review, will be at its discretion. This does not apply to government-directed testing of R&D devices.

3.3 TEST WATER PROPERTIES

3.3.1 Test waters listed in Table 3-1 are to be used throughout testing. These waters are used to carry the test microorganisms, and should not assist the devices in inactivating the microorganisms. At microbiological sampling points, a deionized water (DI) with a conductivity of less than 2 μS/cm shall be used to prepare the test waters. While it is recommended that DI water be used to prepare all test waters, dechlorinated tap water adjusted to the requirements of Table 3-1 may be used between sample points. No disinfectant of any kind shall be present in any of the test waters prior to treatment. Each device is to be tested using two test waters, the General Test Water, and the appropriate Challenge Test Water. The characteristics of the General Test Water and Challenge Test Waters for devices using halogen disinfectants, filtration, ultra-violet light, and copper / silver disinfectants are shown in Table 3-1. Also shown is a test water to be used without microorganisms to assess the leaching of silver into the treated water from silver-impregnated devices.

3.3.2 For technologies not listed in Table 3-1,the General Test Water should be adequate for the unstressed (General) testing phase. For the stressed (Challenge) testing phase, the test sponsor and ETO must determine what physical and chemical parameters should be used to make up a technology-specific Challenge Test Water that will sufficiently challenge the device to ensure that it provides adequate treatment under stressed conditions. The ETO should provide the Challenge Test Water composition to the GRA as part of the PSTP for review and approval. In addition to test water characteristics, variables such as low battery power, environmental conditions (very hot/cold), and other technology-specific efficiency reducers should be employed in the testing procedures and documented in the report.

Table 3-1[1] – Test Water Constituents for Testing SWPs.

Constituent	1 General	2 Halogen Challenge	3 Filter Challenge	4 UV Challenge	5 Cu/Ag Challenge	6 Silver (leaching)
Chlorine[2] (mg/L)	≤ 0.1	≤ 0.1	≤ 0.1	≤ 0.1	≤ 0.1	≤ 0.1
pH	7.0 ± 0.5	9.0 ± 0.2 5.0 ± 0.2[3]	9.0 ± 0.2	7.0 ± 0.5	9.0 ± 0.2	5.0 ± 0.2
TOC (mg/L)	0.1 - 2	10 - 15	10 - 15	10 - 15	12 ± 2	1 ± 0.5
Turbidity (NTU)	< 1	30 - 50	30 - 50	30 - 50	30-50	< 1
Temp (°C)	20 ± 5	4 ± 1	4 ± 1	4 ± 1	4 ± 1	20 ± 5
TDS (mg/L)	50 – 500	1500 ± 300	1500 ± 300	1500 ± 300	1500 ± 300	25 – 100
Alkalinity[5] (mg/L as $CaCO_3$)	100 ± 20	100 ± 20	100 ± 20	100 ± 20	100 ± 20	100 ± 20
Background Bacteria[4] (cfu/100 mL)	$\geq 10^3$	$\geq 10^3$	$\geq 10^3$	$\geq 10^3$	$\geq 10^3$	-
DOC[6] (mg/L)	measure and report	measure and report	measure and report	measure and report	measure and report	-
UVT[7]	-	-	-	measure and report	-	-

[1] Partially assembled from NSF P231; when creating test waters laboratories should target middle of ranges shown.
[2] All chlorine shall be reduced to below detection limits. Chloride in the test water may interfere with chlorine measurements so analysis shall be conducted on the base water prior to adjustment.
[3] For iodine based disinfectants the Challenge Test Water phase will be split between pH 5 and pH 9 as described in the PSTP with at least one microbiological challenge point in each pH.
[4] Bacteria type chosen from those listed in Table 2-1. Refer to Appendix E for estimation of background concentrations in natural waters.
[5] Added as buffer, target value - not subject to test rejection if outside range shown
[6] Analyzed as TOC after filtration through 0.45μm filter. Filter blank must be analyzed to demonstrate organic content attributable to filter.
[7] Ultra-violet light transmittance at 254 nm

3.4 RECOMMENDED MATERIALS FOR ADJUSTING TEST WATER CHARACTERISTICS

- pH: inorganic acids or bases (e.g., HCl, NaOH)
- Total Organic Carbon (TOC): tannic acid
- Turbidity: International Organization for Standardization (ISO) specification 12103-A2 fine test dust.
- Total Dissolved Solids (TDS): sodium chloride, Sigma Chemical Co., S9883 (St. Louis, MO), or other equivalent source of TDS
- Alkalinity: sodium bicarbonate

3.5 PURIFIER SPECIFIC TEST PLAN

3.5.1 The following requirements apply to testing of all SWPs. Additional requirements specific to IWPs and SUWPs are provided in subsequent sections.

3.5.2 The test sponsor, together with the ETO, will produce a written, device-specific PSTP and forward it to the GRA for review and comment prior to beginning testing. The test plan should provide for the evaluation of candidate SWP performance under both normal and stressed operating conditions. It is preferred that the PSTP template provided in Appendix D be used to expedite GRA review and be submitted in Microsoft Word to allow for GRA commenting. Specific questions and PSTP submissions should be directed to the USAPHC Water Supply Management Program at (410) 436-3919, water.supply@amedd.army.mil.

3.5.3 The ETO will maintain the specified test water quality throughout each phase of testing (e.g., General or Challenge Phase), not just at sample points. The laboratory will verify the test water composition using analytical procedures at the beginning of each day, or more often if necessary to ensure compliance with Table 3-1. Additionally, test water turbidity, pH, and temperature measurements will be taken at each sample point, and two other times during each day of testing to ensure compliance with the ranges shown in Table 3-1. On non-sampling days, the stated parameters will be measured three times daily (beginning, middle, end). Details on how the test water was prepared and transferred to the devices for processing shall be included in the final report.

3.5.4 Microbiological challenge organisms will be separated into two sequential challenges separated by a 10 void volume seeding. In order to quantify coliphage type they will be applied in separate challenges.

3.5.5 Testing procedures must replicate field use. Water quality and operating conditions during the stress testing phase using the Challenge Test Water should provide the IWP a realistic worst-case challenge.

3.5.6 In general, devices should be tested without any prior conditioning or seasoning with test water or any other water. Should conditioning be required the PSTP must clearly explain the procedures, including conditioning water quality, and conditioning steps must be provided to the user in the supplied device instructions. Conditioning steps must be executable in field environments. Conditioning will use General Test Water, including background bacteria. Conditioning volume does not accumulate against the test volume.

3.5.7 If the SWP can be operated in more than one way, all methods of operation should be evaluated. Alternatively, if one method can be demonstrated to be the most rigorous of the method options, only that method need be used during testing.

3.5.8 If the SWP uses a cleanable element with a finite number of cleanings (e.g., ceramic element), testing must demonstrate that the device meets the microbiological reduction requirements after the element has been cleaned the maximum number of times allowed in the manufacturer's directions (if applicable).

3.5.9 **C**hallenge solutions containing the microbiological contaminants will be continuously mixed. Influent background bacteria concentrations must be maintained at or above the specified influent concentrations except when the concentrations of all microbial contaminants are increased to the requirements of Table 2-1 at sampling points as described in Section 3.6.4.

3.5.10 Devices that use consumable power sources (e.g., batteries) must incorporate at least one sample point during the final 20% of the power source life. The power source life must be demonstrated during testing. Additionally, treatment systems that incorporate technologies whose effectiveness may vary with time, must treat an amount of water equivalent to the manufacturer-stated capacity of the device or the capacity stated in the PSTP (e.g., a battery operated batch disinfectant device shall be operated sufficient times to treat the entire production capacity stated in the PSTP).

3.5.11 An appropriate sample volume will be retained as a backup for confirmation or retesting purposes when necessary. All pathogens will be analyzed in triplicate with geometric means reported.

3.6 PURIFIER SPECIFIC TEST PLAN – INDIVIDUAL WATER PURIFIER

3.6.1 Devices should generally be operated in batch mode IAW the instructions contained in the commercial package. If continuous-flow testing is preferred by the test sponsor and ETO, the reasoning, proposed plumbing, and test procedures must be specified in the PSTP.

3.6.2 A basic sampling schedule and plan to test the device for short-term use is shown in Table 3-2. Test plans for a specific model may require variations from the plan shown and should be designed to provide worst-case testing.

3.6.3 Minimum test capacity is 135 L unless it can be shown that this is inappropriate for the intended use of the device. Testing should be designed to demonstrate the capacity of the device out to the point where functional component replacement is required.

3.6.4 Prior to each sampling point the device shall be seeded with at least 10 void volumes of test water at the sampling point pathogen concentrations shown in Table 2-1. For ease of testing of batch system, it is recommended that one full batch be used for seeding. At the sampling point a 3 L volume (or full batch, depending on batch size) of sampling point pathogen concentration test water will be processed by the device. The entire processed volume will be collected as the sample and used from which to pull aliquots for individual analyses. For non-batch systems, sample size will be based on the intended use of the device.

3.6.5 For devices that us a filtration step, the final (100%) sampling point will be collected to document the capabilities of the device at the point of clogging. The final sample will be collected at the point of clogging (at or beyond the 100% projected capacity), represented by the reduction in flow as described in the manufacturer instructions for triggering cleaning or a filter change, or if not stated, a 75% reduction of flow as compared to the Day 1 star-up flow rate.

Table 3-2 – IWP Recommended Basic Sampling Plan.

Test Point: Percent of Estimated Capacity[1]	**Microorganism Delivery Media**	**Treatment Chemical Tests[2]**		**Microbiological Tests**	
First Draw	General Test Water	Influent	Effluent	Influent	Effluent
25%	General Test Water	Influent	Effluent	Influent	Effluent
50%[3]	General Test Water	Influent	Effluent	Influent	Effluent
60%	Challenge Test Water	Influent	Effluent	Influent	Effluent
75%	Challenge Test Water	Influent	Effluent	Influent	Effluent
After 48-Hour Rest, First Draw[4]	Device stored at room temperature following manufacturer's recommendations. Challenge Test Water, no pathogen challenge	Influent	Effluent	---[5]	Effluent
100% Clogging Point[6]	Challenge Test Water	Influent	Effluent	Influent	Effluent

[1] Capacity based on minimum 135 L, as described in paragraph 3.6.5.
[2] Treatment Chemical refers to any chemical added to the test water as part of the device's water treatment process (e.g., chemical disinfectant).
[3] Immediately after this sample is collected, test water must change to Challenge Test Water requirements.
[4] If stagnation is not appropriate for the device technology, this becomes a standard sampling point.
[5] The geometric mean of the three previous influent concentrations will be used to calculate the log reduction after stagnation.
[6] The clogging point will be collected in accordance with paragraph 3.6.5.

3.7 PURIFIER SPECIFIC TEST PLAN – SMALL UNIT WATER PURIFIER

3.7.1 The plumbing and recommended points of sampling should be specified in the PSTP.

3.7.2 A basic sampling schedule and plan to test the device for short-term use is shown in Table 3-3. Testing plans for a specific model may require variations from the plan shown in Table 3-3 and should be designed to provide worst-case testing.

3.7.3 The device will be operated for a minimum of 4 hours per day, excluding start-up procedures. Microbial challenges and subsequent sampling will be performed at the end of each sampling day.

3.7.4 Prior to each sampling point the device shall be ”seeded” with at least 10 void volumes, of test water at the elevated microbiological challenge concentrations shown in Table 2-1. At each sampling point sufficient effluent volume for all analyses will be collected as a batch then aliquots drawn for individual analyses.

3.7.5 During the Challenge Test Water phase, the TOC and turbidity of the test water will be increased to meet the requirements of Table 3-1 at sampling points only, including the 10 void volume seeding prior to each sampling point. All other water quality parameters will meet Table 3-1 throughout testing. Seeding volume shall accumulate against the test capacity volume.

3.7.6 For devices that us a filtration step, the final (100%) sampling point will be collected to document the capabilities of the device at the point of clogging. All water beyond 4 hrs of operation on day 10 will contain TOC and turbidity at the challenge values shown in Table 3-1, with the intention of clogging the system. The final sample will be collected at the point of clogging, represented by the reduction in flow as described in the manufacturer instructions for triggering a filter change, or if not stated, a 75% reduction of flow as compared to the Day 1 star-up flow rate.

Table 3-3 – SUWP Recommended Basic Sampling Plan.

Test Points[1]	Test Water Quality	Test Water Notes	Treatment Chemical[2]		Microbiological Tests	
First Draw, Day 1	General Test Water	-	Influent	Effluent	Influent	Effluent
Day 3	General Test Water		Influent	Effluent	Influent	Effluent
Day 5[3]	General Test Water		Influent	Effluent	Influent	Effluent
Day 6	Challenge Test Water	TOC and turbidity added only at sampling points.	Influent	Effluent	Influent	Effluent
Day 8	Challenge Test Water		Influent	Effluent	Influent	Effluent
After 48-Hour Rest, First Draw[4]	Device stored at room temperature following manufacturer's recommendations. Challenge Test Water, no pathogen challenge		Influent	Effluent	---[5]	Effluent
Day 10 / Clogging Point[6]	Challenge Test Water	All test water after Day 10 will include TOC and turbidity.	Influent	Effluent	Influent	Effluent

[1] Based on 4 hours of flow/day (including start up).
[2] Treatment Chemical refers to any chemical added to the test water as part of the device's water treatment process (e.g., chemical disinfectant).
[3] Immediately after this sample is collected, test water must change to Challenge Test Water requirements.
[4] If stagnation is not appropriate for the device technology, this becomes a standard sampling point.
[5] The geometric mean of the three previous influent concentrations will be used to calculate the log reduction after stagnation.
[6] The clogging point will be collected in accordance with paragraph 3.7.6.

3.8 ANALYTICAL PROCEDURES

3.8.1 General

3.8.1.1 Both treatment chemical (effluent) and microbiological (influent and effluent) sampling and analysis shall be conducted at each test point indicated in Table 3-2 or as specified in the approved PSTP. All physical and chemical analyses shall be conducted IAW procedures in the current version of *Standard Methods for the Examination of Water and Wastewater* (reference 7) or equivalent.

3.8.1.2 Quality assurance testing shall be conducted for the seed microbes under environmental conditions of the test (pH, turbidity, temperature, TDS and TOC) to make sure that there is no significant change over the test period.

3.8.1.3 When the IWP employs a disinfectant (e.g., halogen or silver), the active agent residual will be measured in the effluent at each microbiological test point. Immediately after collection (allowing for the contact time specified in the manufacturer's instructions for the product water), each test sample must be treated to neutralize any residual disinfectant. For silver-based disinfectants, thioglycollate-thiosulfate neutralizer solution (reference 8) is appropriate, and for halogens, sodium thiosulfate shall be used. This solution should be prepared daily. All results are invalid unless samples are neutralized, and neutralization is verified, immediately upon collection.

3.8.1.4 In addition to the basic sampling plan, when the devices being tested contain silver, two new devices will be tested using the silver leaching test water. Testing will follow the same testing procedures, but without the microbiological loading, to ensure the silver is not leached into the effluent at concentrations exceeding the short term MEG for silver specified in the current edition of the USAPHC TG 230 (reference 6), or the corresponding USEPA NPDWR MCL if there is no MEG. The sample plan will be based on procedures described in the USEPA Guide Standard (reference 1). If a different bacteriostatic material is used within the device, procedures shall be developed to evaluate leaching of that material.

3.8.2 Microbiological Methods

The following section describes the microorganisms to be used by the ETO to demonstrate the reduction capabilities of the IWP. There are three categories of microorganisms: bacteria, viruses, and protozoa. These microorganisms are representative of the types of microbiological contaminants expected in untreated source waters and to which the SWP might be applied.

3.8.2.1 Selection of Microorganisms

3.8.2.1.1 The microorganisms selected for inclusion in this Protocol are well documented as laboratory test organisms. Some are also pathogenic organisms in and of themselves and, therefore, have an added significance if the SWPs demonstrate successful reduction. They have varying degrees of susceptibility to commonly used drinking water disinfectants. Additionally, the microorganisms represent an array of particle sizes that should provide useful information with respect to SWPs that rely on mechanical size exclusion for the reduction of microbes.

3.8.2.1.2 The bacteria species *Raoultella terrigena* (*R. terrigena*, formerly *Klebsiella terrigena*), *Escherichia coli* (*E. coli*), or *Bacillus atrophaeus (B. atrophaeus,* spore form*)* may be used to represent the challenge of bacterial contaminants. The ETO will select one of the bacteria species for test purposes. These three bacterial genera, as well as members of the Enterobacteriaceae family, have a history of use in disinfection studies and protocols (references 1, 9 and 10). *R. terrigena* and *E. coli* are typical of the total coliform bacteria group frequently found in untreated surface waters. The coliform bacteria levels in many rivers and streams with typical pollution sources have been found to be 10^5/100 mL or higher. *E. coli* has added health significance as its presence is very indicative of fecal contamination. Some strains of *E. coli* produce a strong toxin that can lead to severe gastrointestinal illness. *B. atrophaeus*, a species of the *Bacillus* genus, shares many characteristics with *Bacillus anthracis*, the organism that causes anthrax.

3.8.2.1.3 In the virus category, MS2 and fr bacteriophages or "coliphages" are the viruses recommended for testing of the SWPs. MS2 is a virus that infects *E. coli* bacteria and has been well documented in literature as a challenge organism for disinfection studies. Enteric viruses are responsible for many waterborne illnesses. Studies have indicated that enteric virus concentrations in typical polluted waters approach 10^2 per liter of water (reference 1). These phages were chosen as surrogates based on their small size and isoelectric points. fr is 19 nm in diameter with an isoelectric point at pH 8.9, and MS2 is 24 nm in diameter with an isoelectric point at pH 3.9. The small diameter of both of these viruses approximates the size of many of the smaller enteric viruses. The isoelectric point is the pH at which the virus is neutrally charged. The viruses have varying isoelectric points, so they will have different surface charges, or different strengths of negative or positive charge at the different pH values. In solutions with pH above the isoelectric point, the virus will be negatively charged; below it, the virus will be positively charged. fr Phage should be used with MS2 for the viral challenge due to its small size and high isoelectric point. This approach serves to evaluate whether electrostatic forces play a role in virus retention in addition to mechanical filtration.

3.8.2.1.4 *Cryptosporidium parvum* is a very common protozoan found in the majority of surface water supplies in the U.S. It causes Cryptosporidiosis, a severe gastrointestinal illness. Common disinfection practices used in public water systems are not effective against *Cryptosporidium*. Therefore, municipal water systems using surface waters are required to reduce *Cryptosporidium* prior to disinfection, generally by adding or improving filtration operations.
In the environment, the organism exists in a protective cyst stage called an oocyst. *Cryptosporidium* oocysts are typically 3-5 microns in diameter. For purposes of testing the SWPs, *C.* parvum is recommended over *Giardia lamblia*, which was the protozoan used in the

USEPA Guide Standard (reference 1). *Cryptosporidium* oocysts are smaller in size compared to the *Giardia* cysts (which are 8-10 microns in diameter) and *Cryptosporidium* is much more disinfectant-resistant than *Giardia*.

3.8.2.1.5 Bacteria, viruses, and phages can be obtained from the American Type Culture Collection (ATCC), Manassas, VA or an equivalent source. *Cryptosporidium* may be obtained from Waterborne Inc., New Orleans, LA, Bunch Grass Farms in Idaho, or an equivalent source. Additional contact information for these sources is provided below. Sourcing is not limited to those listed in this document as these are provided only as examples. There may be other legitimate commercial sources for the purchase of the microbial cultures. The following criteria are essential for demonstrating equivalency:

- use of the same isolate strain
- use of the same host species
- use of same processing and cleanup techniques
- demonstration of comparable ID50 value
- For example, if another vendor is used to obtain *Cryptosporidium* oocysts, ensure the use of *Cryptosporidium parvum*, Iowa isolate, or a strain identified as equivalent

3.8.2.2 Bacterial Tests

3.8.2.2.1 Chosen Organism(s). Select one of the following bacterial species for the bacterial challenge: *Raoultella terrigena* (ATCC 33257), *E. coli* (ATCC 11229), or *Bacillus atrophaeus* (ATCC 9372).

3.8.2.2.2 Method(s) of Production. Suggested methods follow, but all methods that have been validated as equivalent (i.e., regarding challenge organism growth phase, arrangement and morphology) are acceptable.

a. *Raoultella terrigena* (ATCC 33257) and *E. coli* (ATCC 11229)

- The organism will be provided in the lyophilized form by American Type Culture Collection (ATCC) or equivalent. The lyophilized culture will be rehydrated according to the supplier's instruction.
- A fresh batch of organism must be prepared for each day of use.
- To prepare, transfer 1 mL of the challenge organism stock to 100 mL of Tryptic Soy Broth (TSB), place on a rotary shaker set to 150 RPM, and incubate at 35 ºC for 16-18 hours. Following incubation, the organisms will be harvested by centrifugation at 2,500 x g for 10 minutes. The pellet will be resuspended in 50 mL PBDW and the washing procedure repeated two additional times.
- The density of the suspension can be confirmed by optical density (OD) measurement at 415 nm. Prior to confirming by OD_{415}, each laboratory shall

initially perform a calibration curve that correlates plate counts to OD_{415} and to verify late log phase growth.

- Use a sufficient volume of TSB to provide the correct volume and concentration of the organism to challenge the test units based on flow rate, capacity, etc.

b. *Bacillus atrophaeus* (ATCC 9372) must be prepared in a spore suspension for delivery to the test system. Spore suspensions may be prepared as follows:

(1) **Method A** (using previously harvested *B. atrophaeus* spores)

(a) Aseptically inoculate, by streak plating technique, several tryptic soy agar (TSA), petri plates (100 x 15 mm).

(b) Incubate for 48 ± 2 h at 99 ± 1 °F (37 ± 0.5 °C).

(c) Remove characteristic (pigmented dark orange) colonies and transfer to ten 220 mL sterile screw-capped bottles each containing approximately 50 mL of TSA.

(d) Incubate for 48 ± 2 h at 99 ± 1 °F (37 ± 0.5 °C).

(e) Add 10 mL of phosphate buffered saline (PBS) to each slant and gently wash the bacteria from the agar surface.

(f) Combine the bacterial suspensions to yield approximately 100 mL in a sterile 150mL screw-cap bottle. Heat shock culture at 149 ± 1 °F (65.0 ± 0.5 °C) for 15 min.

(g) Determine spore concentration by standard dilution-plate methods using PBS and TSA. Spores prepared as above should yield an average count of 2×10^9 to 4×10^9 cfu/mL.

(h) Incubate plates for 48 ± 2 h at 99 ± 1 °F (37 ± 0.5 °C).

(i) Store the stock spore suspension (2×10^9 to 4×10^9 cfu/mL) at 39 °F (4 °C) or divide into aliquots to store in screw-capped vials at -94 °F (-70 °C).

(2) **Method B**

(a) Inoculate 250 mL portions of sterile tryptose broth with aliquots of previously harvested *B. atrophaeus* spores; or rehydrated freeze-dried cultures per ATCC or the National Culture Type Collection (NCTC) instructions.

(b) Incubate on a reciprocating shaker for 48 ± 2 h at 99 ± 1 °F (37 ± 0.5 °C).

(c) Heat shock cultures at 149 ± 1 °F (65 ± 0.5 °C) for 15 min.

(d) Transfer suspensions to screw-cap test tubes and wash at least three times in sterile distilled water by centrifugation at 2500 rpm for 15 min. Use PBS in last washing if storage is required.

(e) Determine spore concentration by standard dilution-plate methods using PBS and TSA. Spores prepared as described above should average 1.5 x 10^9 cfu/mL.

(f) Incubate plates for 48 ± 2 h at 99 ± 1 °F (37 ± 0.5 °C).

(g) Store the stock spore suspension (2 x 10^9 to 4 x 10^9/mL) at 39 °F (4 °C) or divide into aliquots to store in screw-capped vials at -94 °F (-70 °C).

3.8.2.2.3 State of the Organisms. Organisms in the late log phase/early stationary phase of growth and suspended in phosphate buffered saline will be used. The growth phase of the cultures need to be confirmed via in-line biomass monitoring and/or direct cell counts every time the cultures are prepared. Refer to Appendix E for a growth curve of *E. coli* and *R. terrigena.*

3.8.2.2.4 Assay Techniques for Stock, Influent and Effluent Samples. Assays to determine challenge quantities of the organism may be by the pour plate, spread plate, or membrane filter method using TSA, or equivalent media, for *B. atrophaeus* and m-Endo agar, or equivalent media, for *R. terrigena* and *E. coli* (see Section 9215, Standard Methods, reference 7). Assay each sample dilution in triplicate.

3.8.2.2.5 Example Culture Sources.

ATCC
P.O. Box 1549
Manassas, VA 20108
800-638-6597, Fax: 703-365-2750
www.atcc.org

Presque Isle Cultures
P.O. Box 8191
Erie, PA 16505
814-833-6262, Fax: 814-833-2834
www.picultures.com

3.8.2.3 Virus Tests

3.8.2.3.1 Chosen Organism(s). MS2 coliphage (ATCC 15597-B1) and host *Escherichia coli* (Migula) Castellani and Chalmers ATCC 15597, and fr (ATCC 15767-B1) and host *Escherichia coli* (Migula) Castellani and Chalmers ATCC 19853.

3.8.2.3.2 Methods of Production. The bacteriophage suspensions can be cultivated in-house or purchased from several commercial vendors. All methods that can demonstrate the cultures are monodispersed are acceptable. One acceptable method for in-house cultivation, is to make serial dilutions from an existing coliphage stock suspension so that plates possessing a confluent field of plaques are obtained through the top agar plating method. The top agar plating method follows:

a. 4 to 18 hours prior to plating, place 100 μL of a 1×10^8 cfu/mL appropriate host *E. coli* suspension in a fresh 10 mL aliquot of TSB and incubate at 35 °C.

b. Transfer 100 μL of the resulting *E. coli* culture to a sterile test tube.

c. Serial dilute the stock bacteriophage culture in PBDW.

d. Transfer 1 mL of the dilutions into the tubes containing the appropriate host *E. coli*.

e. Vortex the samples for a minimum of 30 seconds to "mate" the bacteria and bacteriophage.

f. Add 4 mL of molten, tempered 1% TSA to each tube.

g. Pour the agar/organism mixture evenly over a labeled TSA plate and rotate to ensure even coverage over the plate.

h. Allow the plates to solidify and then incubate at 35 °C for 18-24 hours.

i. Observe the viral plaques using a colony counter.

j. Use the plates possessing confluent plaques for viral extraction. To achieve 50 mL of a 1×10^{10} pfu/mL coliphage stock suspension, 10-15 plates must be processed.

k. Add 5 mL of PBDW directly to the surface of each plate.

l. Allow the plates to sit at room temperature for 1 hour while the bacteriophages diffuse from the agar into the overlaying PBDW.

m. Harvest the 5 mL of PBDW suspension and combined with the suspension from the other plates.

n. Place the suspension in a sterile 250 mL centrifuge bottle and centrifuge at 2,500 x g to remove cellular debris.

o. Save the supernatant and filter through a sterile Whatman® #2 filter paper using a vacuum supply and sterile filter flask. It may be necessary to use several filters if clogging becomes excessive. Further filter through a 47 mm 0.45 μm membrane filter, and then through a 0.1 μm filter for purification.

® Whatman is a registered trademark of Whatman International Ltd., 27 Great West Road, Brentford, Middlesex, TW8 9BW, United Kingdom.

p. Determine the final concentration of bacteriophage using the top agar overlay technique at appropriate dilutions (expect 10^9 to 10^{11} pfu/mL). Incubate plates at 35 °C for 24 ± 2 hours prior to reading.

q. This suspension can be stored at 2-4 ºC indefinitely. Enumeration, however, should be performed less than 1 week prior to each use to ensure proper concentration.

Alternatively, the coliphage stock suspensions may be purchased from a qualified subcontract laboratory who employs the aforementioned filtration/purification scheme.

3.8.2.3.3 State of the Organisms. MS2 and fr will be provided in a freeze-dried state. Specific guidance to rehydrate and grow the virus in *E. coli* will be provided by the supplier.

3.8.2.3.4 Assay Techniques. The double agar (top agar overlay method) or single agar layer (SAL) procedure may be used. See guidance above and contained in USEPA Method 1601, "Male-specific (F+) and Somatic Coliphage in Water by Two-step Enrichment Procedure" (reference 14) and USEPA Method 1602, "Male Specific and Somatic Coliphage in Water by SAL." Additional information about virus assays may be found in Method 9510G, "Assay and Identification of Viruses in Sample Concentrates," in Standard Methods (reference 7) and in Adams, M.H., 1959. *Bacteriophages*, Interscience Publishers, New York (reference 18). Assay each dilution in triplicate.

3.8.2.3.5 Example Culture Source.

ATCC
P.O. Box 1549
Manassas, VA 20108
800-638-6597, Fax: 703-365-2750
www.atcc.org

3.8.2.4 Oocyst Tests

3.8.2.4.1 Chosen Organism *Cryptosporidium parvum.*

3.8.2.4.2 Method of Production. Iowa isolate, produced from experimentally infected cows.

3.8.2.4.3 State of Organism. Live oocyst specimens are prepared in phosphate-buffered saline solution and treated with antibiotics, unless otherwise requested. Typical aliquots are available in 1 x 10^6 (1 million) and 5 x 10^6 (5 million). The live oocysts are guaranteed to remain at least 50% viable for 2 months from harvest date.

3.8.2.4.4 Cell Wall Integrity Assay (for all treatment device types). The methods described in NSF Standard 53 (reference 15) for excystation, or equivalent, shall be used to assess the cell wall integrity of the oocyst stock. Greater than 50% excystation must be achieved. Note that this is not an indication of viability. For mechanical filtration devices in which membrane integrity is the key cellular factor in organism performance, excystation may be used as the sole qualifying acceptance criteria for a given stock (e.g., no viability bioassay is required).

3.8.2.4.5 Solely Non-mechanical Filtration Treatment Devices. The infectivity of the cyst stock must be determined by cell culture method (reference 19), or equivalent alternative method, prior to testing.

3.8.2.4.6 Example Culture Sources.

Bunchgrass Farm
1301 Drury Rd
Deary, ID 83823
Ph - 208-877-1276
Fax-208-877-1290
bunchgrassfarm@gmail.com

Waterborne, Inc.
6047 Hurst St
New Orleans, LA 70811
866-895-3338, Fax: 504-895-3338
www.waterborneinc.com

3.8.2.4.7 Sample Processing and Determination of Oocyst Concentration. All oocyst stocks, influent samples and effluent samples shall be processed and enumerated according to the membrane filtration/fluorescent microscopy protocol described in the most current version of NSF Standard 53 (reference 15). The quality control requirements for the oocyst enumeration method described in Standard 53 must also be adhered to**.** All standards are subject to revision; the most recent edition applies.

3.8.2.5 Alternate Use of Spheres. For SWPs that rely on physical filtration processes, testing for the required log reduction of polystyrene microspheres may be used to demonstrate the ability to reduce *Cryptosporidium* oocysts. The polystyrene microspheres shall have 95% of particles in the range of 3.00 ± 0.15 µm. The size variation of the polystyrene microspheres shall be confirmed by electron microscopy. The spheres shall have a surface charge content of less than 2 µEq/g. If a variation is made in the microspheres' specification, the ETO will be responsible for identifying a source of spheres and verifying acceptability for purposes of simulating *Cryptosporidium* oocysts. All stock, influent, and effluent samples shall be processed and enumerated according to Section 3.8.2.4.7.

3.8.3 Records. All pertinent procedures and data shall be recorded in a standard format and retained for possible review until the report of results has been completely accepted by review authorities, in no case for less than 1 year.

3.8.3.1 Log Reduction Calculation. Testing will be conducted simultaneously on three identical devices, termed replicates. At each sampling point, influent and effluent water samples will be collected and each analyzed in triplicate. Log reductions for the purpose of compliance with this Protocol will be calculated at each sample point as follows:

3.8.3.1.1 The geometric mean (GM) of each triplicate analysis (X) will be calculated for each influent and replicate device effluent as:

$$[1] \qquad GM = (X_{1*}X_{2*}\ldots X_n)^{(1/n)}$$

3.8.3.1.2 The log reduction for each replicate at each sample point will be calculated using the results from [1], shown below as the negative $\log_{10}$ of the GM of each replicate effluent, GM_{eff}, divided by the GM of the influent, GM_{inf}.

$$[2] \qquad \text{Log Reduction} = -\log_{10}(GM_{eff}/GM_{inf})$$

3.8.3.2 Acceptance of Records. IAW Section 3.5.3 of the USEPA Guide Standard, three production units (replicates) of the SWP must continuously meet or exceed the log reduction requirements shown in Table 2-1, except that up to 10% of influent/effluent sample pairs may vary from the reductions required in Table 2-1 by:

Viruses:	1 log
Bacteria:	1 log
Cysts:	½ log

In all cases, the GM of all microbiological reductions must meet or exceed the requirements of Table 2-1. Devices that use a single surrogate microbe (e.g., MS2 for UV devices) will not be granted this variation from the required log reduction.

3.8.3.3 Scaling. If a manufacturer produces more than one device using identical technology with similar design and construction, it may be appropriate to allow for acceptance of the performance of one device size as representative of the performance of other devices sizes. In this case the GRA will make the determination as to the appropriateness of accepting data for protocol compliance. In all cases the general rule will allow for acceptance when scaling up to 3 times and down to 1/3 of the tested device.

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SECTION 4
MATERIALS

4.1 Materials in contact with drinking water shall meet the requirements specified under one of the following appropriate NSF/ANSI drinking water standards:

- Manufacturer shall provide a complete wetted parts list;
- Manufacturer shall submit void volumes and system flow information;
- Water contact materials certified to one of the following NSF Food Equipment/Drinking Water Standards: NSF/ANSI 42, 44, 53, 55, 58, 62, 60, 61, and 51 shall be accepted without additional testing;
- A minimum test battery for all systems with non-Certified wetted parts shall include: VOCs, regulated metals, and a BNA scan with the exposure scenario being determined by Toxicology;
- All non-Certified wetted contact materials shall undergo a toxicology review for determination whether any testing in addition to the minimum test battery shall be required; and material extraction testing shall not include "without media" samples. Required test samples shall be "with media" only.

All standards are subject to revision; the most recent edition applies. If the technology of the system does not fall under the scope of one of the above standards, then a comparable evaluation method with equivalent criteria shall be applied to the system.

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SECTION 5
DESIGN AND CONSTRUCTION

5.1 Complete systems shall be watertight and during performance testing operation shall not leak.

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SECTION 6
PRODUCT LITERATURE

6.1 Information setting forth complete, detailed instructions for installation, operation, and maintenance shall be provided with each system. Specific information shall include:

- complete name, address, and telephone number of manufacturer
- model number and trade designation
- flushing and conditioning procedures
- operation and maintenance requirements (including user responsibility, parts, and service)
- statement confirming that the system meets the requirements of NSF Protocol P248 for Military Operations Microbiological Water Purifiers

6.2 Replacement components packaging shall be labeled with the following information (where appropriate):

- model number and name of component
- model number of system(s) in which the component is to be used
- name and address of manufacturer
- rated capacity/rated service life in liters (gallons)
- disinfection and cleaning instructions
- component rated capacity/rated service life in L (gal)
- operating or exchange steps
- required replacement intervals of ultraviolet lamp(s) IAW the manufacturer's instructions (UV systems)

6.3 A performance data sheet shall be available to potential buyers for each system and shall include the following information:

- complete name, address, and telephone number of manufacturer
- model number and trade designation
- statement that the system conforms to NSF Protocol P248 for Military Operations Microbiological Water Purifiers
- statement that the system is not intended to convert wastewater or raw sewage into drinking water
- rated capacity/rated service life in liters (gallons)

NOTE – Each unique model designation shall not claim a capacity or service life greater than the least reduction capacity or service life that has been verified through testing to NSF Protocol P248

- general operation and maintenance requirements including, but not limited to:
  - suggested frequency of component change or service to system
  - user responsibility
  - statement that while testing was performed under standard laboratory conditions, actual performance may vary
- model number of replacement component
- electrical requirements
- explanation of how the performance indicator functions, if present

SECTION 7
PRODUCT LABELING

7.1 ***Product labeling*** shall be in accordance with NSF Certification Policies for NSF Protocols and Non-NSF Standards. Certified individual products or packaging materials should bear the following NSF Mark design:

NSF®

The [INSERT TRADE DESIGNATION] system is certified by NSF International against NSF P248 – Emergency Military Operations Microbiological Water Purifiers

7.2 NSF and the mark are registered trademarks of NSF International. No company or person shall apply or use the NSF Mark in connection with a product or represent in any way that the product is certified until written authorization is provided by NSF. NSF may pursue legal recourse if the mark is misused.

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SECTION 8
AUDITING AND RE-EVALUATION

8.1 Audits of all production locations shall be conducted by NSF at least annually. Audits will cover the following basic items:

- No changes in labeling or product literature
- No change in design, construction or materials without prior approval by NSF
- Recordkeeping

8.2 Re-evaluation of certified systems shall occur every 5 years.

REFERENCES

1. U.S. Environmental Protection Agency (USEPA), “Guide Standard and Protocol for Testing Microbiological Water Purifiers,” Office of Pesticide Programs, Criteria and Standards Division, Office of Drinking Water, Washington, DC, April, 1987.

2. NSF International (NSF), “Protocol for Equipment Verification Testing for Inactivation of Microbiological Contaminants,” Ann Arbor, MI, January, 2003.

3. NSF International, “Microbiological Water Purifiers,” NSF Protocol P231, Ann Arbor, MI, February 2003.

4. International Organization for Standardization / International Electrotechnical Commission (ISO/IEC) 17025:2005, “General requirements for the competence of testing and calibration laboratories.”

5. Soldier Enhancement Program, “Operational Requirements Document for a Water Purification Device (WPD),” ACAT III, 25 July 2003.

6. U.S. Army Public Health Command, “Technical Guide 230: Environmental Health Risk Assessment and Chemical Exposure Guidelines for Deployed Military Personnel,” Aberdeen Proving Ground, MD, 2010.

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APPENDIX A
GLOSSARY OF ABBREVIATIONS AND TERMS

Abbreviations

ANSI	American National Standards Institute
ATCC	American Type Culture Collection
CFU	Colony-Forming Units
COTS	Commercial-off-the-shelf
DOC	Dissolved Organic Carbon
DOD	Department of Defense
ETO	Equipment Testing Organization
EU	European Union
GEMS	Global Environmental Monitoring System
GM	Geometric Mean
GRA	Government Review Agency
IAW	In accordance with
ID	Infectious Dose
ISO	International Organization for Standardization
IWP	Individual Water Purifier
MCL	Maximum Contaminant Level
MEG	Military Exposure Guideline
NSF	NSF International
NCTC	National Culture Type Collection
NPDWR	National Primary Drinking Water Regulations
NTU	Nephelometric Turbidity Units
OD	Optical Density
POC	Point-of-contact
PBDW	Phosphate Buffered Deionized Water
PBS	Phosphate Buffered Saline
PE	Performance Evaluation
PFU	Plaque-Forming Units
pH	Negative Log of Hydrogen Ion Concentration
PSTP	Purifier Specific Test Plan
QA	Quality Assurance
QC	Quality Control
R&D	Research and Development
RPM	Revolutions Per Minute
SAL	Single Agar Layer
SUWP	Small Unit Water Purifier
SWP	Small Water Purifier
TDS	Total Dissolved Solids
TG	Technical Guide
TOC	Total Organic Carbon
TDS	Total Dissolved Solids

TSA	Tryptic Soy Agar
TSB	Tryptic Soy Broth
USAPHC	U.S. Army Public Health Command
USEPA	U.S. Environmental Protection Agency
USGS	U.S. Geological Survey
UV	Ultra-violet
WHO	World Health Organization

Terms

Equipment Testing Organization (ETO) – An organization appropriately certified by federal and/or state accrediting organizations and qualified to conduct independent studies and testing of SWPs IAW protocols and test plans. The role of the ETO is to assist with developing the PSTP in collaboration with the test sponsor, conduct all performance tests, and report results to the designated government review agency.

Individual Water Purifier (IWP) – A device capable of inactivating and/or reducing waterborne pathogens from water that can be carried as part of the Warfighter's pack and utilized without interfering unduly with performance of his mission. To qualify for consideration for military use, a microbiological water purifier must treat or reduce all types of challenge organisms, or be capable of being combined with other devices, to meet specified standards.

Protocol – A written document of procedures for testing that applies to all makes and models of SWPs.

Purifier Specific Test Plan (PSTP) – A written document of procedures for medical efficacy and reliability testing that apply to a specific make and model of small water purifier.

Report – A written document that includes data, test results, findings, and any pertinent information collected IAW the Protocol and the PSTP.

Small Unit Water Purifier (SUWP) – a device/system capable of inactivating/or reducing waterborne pathogens from water that can be used by a group (unit) of military personnel to provide a microbiologically-safe supply of water for consumption and hygiene purposes.

Small Water Purifier (SWP) – a generic term used to describe a device/system that is capable of inactivating and/or reducing waterborne pathogens from water and is smaller than the bulk water systems currently fielded by the U.S. Military.

Test Sponsor – A manufacturer, vendor, non-government, or government organization wanting to test small water purifiers. The role of the test sponsor is to supply, devices, fund, and collaborate with the ETO in preparation and execution of the PSTP.

APPENDIX B
GOVERNMENT REVIEW AGENCY COMPLIANCE

1.1 GRA Protocol P248 Compliance

1.2 The GRA reserves the right to determine compliance with this protocol for military purposes. Only manufacturers whose devices have been tested, whose data have been reviewed by the GRA, and who have received a GRA P248 Letter of Compliance, may market those devices as having met the GRA requirements of this protocol. The GRA may grant compliance with NSF Protocol P248 Appendix B. The GRA does not review for, and may not grant the use of, the NSF Mark, which may only be granted by NSF International.

1.3 The GRA's involvement with this Protocol is to ensure that SWPs that may be used by U.S. Military Warfighters undergo a thorough evaluation of device capabilities with respect to reducing the potential for ingestion of waterborne pathogens. The performance requirements, as described in Section 2, are deliberately challenging so as to demonstrate device capabilities under stressed conditions and discriminate between the treatment capabilities different SWPs provide for each required class of pathogen. The goal is to provide Servicemembers with effective treatment options regardless of whether the devices are provided to Warfighters by the U.S. Military or are produced and marketed as COTS items.

1.4 The following describes the minimum requirements that must be met for the GRA to grant NSF P248 compliance.

a. The PSTP must be reviewed and accepted by the GRA prior to testing initiation.

b. Testing must be conducted by a GRA-approved third-party laboratory to the requirements set forth in the approved PSTP.

c. The full laboratory testing results report must be sent to the GRA for review.

d. A description of the materials used within, and to manufacture, the device must be provided to the GRA, along with information as to the safety of the materials for contact with drinking water. This includes information on any active agents used as part of the treatment process.

If these steps are followed, the GRA will provide the testing initiator with a letter stating whether the results and material safety check warrant device compliance with Protocol P248 Appendix B. If compliance is not granted, an explanation of why will be provided.

2.1 GRA Protocol P248 Partial Compliance

2.2 Acknowledging the possibility that several treatment technologies, potentially from multiple devices, may be required to meet all of the Protocol performance requirements of Section 2, the GRA may approve devices for conformance with only the requirements it meets (e.g., a class of pathogen). This partial P248 compliance is meant to clearly provide potential users with the capabilities of each device so as to promote informed decisions when choosing a device, or device combination, that meets mission and health risk reduction requirements.

2.3 The GRA P248 Letter of Compliance provided to the testing sponsor will include a compliance/non-compliance rating for each protocol performance requirement. Partial protocol compliance is solely meant to provide potential users with information on device capabilities and should not be confused with, misrepresented, or marketed as full Protocol P248 compliance.

2.4 Protocol requirements ensure that all devices are tested against sufficient pathogens and/or surrogates to determine device capabilities and as such, testing must include challenging devices with all applicable and specified organisms. Partial P248 Compliance does not change this requirement. Devices that are marketed as having met portions of Protocol P248 must describe, as stated in the letter provided by the GRA, the requirements the device met and those requirements that it did not meet so as to fully disclose device capabilities.

2.5 The U.S. Army Public Health Command – Water Supply Management Program may be contacted at (410) 436-3919, water.supply@amedd.army.mil for clarification.

APPENDIX C
SUGGESTED OUTLINE FOR A PURIFIER SPECIFIC TEST PLAN

TITLE PAGE

EXECUTIVE SUMMARY - The Executive Summary describes the contents of the PSTP (not to exceed two pages). A general description of the equipment shall be included, as well as the testing location, a schedule, and a list of participants.

TABLE OF CONTENTS

1. PURPOSE

2. INDIVIDUAL WATER PURIFIER CAPABILITIES AND DESCRIPTION. The PSTP shall include:

a. Description of the candidate IWP system, including all component parts and functions, to include photographs of the physical system from relevant angles or perspectives. The description of the equipment shall include at least the following information:

(1) Equipment name
(2) Model #
(3) Manufacturer name and address
(4) Principles of operation and pathogen reduction
(5) Size and weight
(6) Estimated production capacity and cleaning procedures
(7) Logistical requirements

b. An introduction and discussion of the engineering and scientific concepts on which the microbiological inactivation capabilities of the candidate IWP are based.

c. A description of the equipment and each process included as a component in the modular system including all relevant schematics.

d. A brief description of the physical construction/components of the IWP, including the general notable limitations, required consumables, ruggedness, etc.

e. A statement of typical rates of consumption of chemicals.

3. ETO NAME AND CREDENTIALS

4. DEVICE TESTING PROCEDURES. The PSTP shall include the following:

a. Detailed conditioning and operating procedures for the IWP with listing of operating parameters.

b. Test water characteristics, application, and frequency of analysis of waters for ensuring compliance with required water characteristics.

c. Microbial challenge, processing, and enumeration procedures.

d. Sampling and analysis schedule for water quality and microbial parameters.

e. Requirements for initiating device cleaning and test completion.

f. Procedures for determination of effective use lifetime.

g. Description of the statistical design, QA/QC procedures and statistical techniques that will be used to analyze the data.

h. Identification of additional parameters of treated water quality and analytical methods that will be employed.

i. Description of the methodologies to be used for conducting the microbiological inactivation challenge studies.

j. Outline of the testing schedule.

5. ROLES AND RESPONSIBILITIES. The PSTP shall include roles and responsibilities of all involved with testing. This should specifically address the test sponsor and their relationship with the ETO as described below.

a. The test sponsor shall be responsible for the following:

(1) Provision of new, commercially packaged equipment for testing complete with instructions.

(2) Provision of logistical and technical assistance to the ETO, as required.

(3) Provision of a complete list of all equipment and materials to be used in the verification testing.

(4) Provision of field operating procedures.

b. The ETO is responsible for the following:

(1) Assisting the test sponsor in developing the PSTP and conducting efficacy testing IAW the PSTP.

(2) Providing needed logistical support.

(3) Establishing a communication network.

(4) Scheduling and coordinating the activities of all testing participants.

(5) Managing, evaluating, interpreting, and reporting on all test results; and evaluating and reporting on the performance of the IWPs.

c. The ETO shall further be responsible for including the following elements in the PSTP:

(1) A definition of the roles and responsibilities of appropriate medical efficacy testing.

(2) A table which includes the name, affiliation, and mailing address of each participant, a POC, description of participant's role, telephone and fax numbers, and e-mail addresses.

(3) A schematic of the organizational structure of operational and analytical support.

d. The GRA shall be responsible for clarifying this Protocol to test sponsors and ETOs as required, and for reviewing and approving the PSTP for each IWP offered by the vendor for use in the U. S. Military in a timely manner.

APPENDIX A – REFERENCES.

APPENDIX B – ABBREVIATIONS AND ACRONYMS.

APPENDIX C – QUALITY ASSURANCE PROJECT PLAN. The responsibility for ensuring that IWP operation, sampling, and analysis activities comply with the QA/QC requirements of the PSTP shall rest with the ETO. QA/QC activities for a laboratory that analyzes samples sent off-site shall be the responsibility of that analytical laboratory's supervisor. The PSTP shall include:

a. Description of methodology for measurement of accuracy and precision.

b. Description of the methodology for use of blanks, the materials used, the frequency.

c. Criteria for acceptable method blanks and the actions if criteria are not met.

d. Description of procedures to be used for determination of microbial viability.

e. Definition of data to be reported during the Testing Program, in terms of analytical parameter type and frequency.

f. Description of all techniques to establish (where applicable) the representativeness completeness, accuracy and precision of methods in the analysis of water quality parameters and microbiological contaminants.

APPENDIX D – HEALTH AND SAFETY PLAN. The PSTP shall address safety considerations that are appropriate for the equipment being tested and for the challenge organisms, if any, being used in the verification testing.

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APPENDIX D
ESTIMATION OF THE BACTERIA CONCENTRATION IN NATURAL WATERS

1.1 Background

1.1.1 The bacteria challenge concentration of NSF Protocol P248 is higher than what is expected to be present in natural waters. This elevated "challenge" concentration [10^7 colony forming units (cfu)/100mL] is necessary during sampling events to show SWP bacterial log reduction capability. Since the goal of NSF P248 is to test SWPs under conditions that represent field use, a bacterial "background" concentration is needed for use between sampling events to more closely represent natural water conditions.

1.1.2 The bacterial quality of natural waters varies with location as well as with changing environmental conditions at each location. The comparison of the relative bacterial qualities of waters at different locations is based on the detection of indicator organisms. Total and fecal coliforms are the most commonly used indicator organisms and those stated in regulations for determination water use classification (Table D-1). The "background" concentration of these indicator organisms gives a good estimate of the pathogenic bacteria present in natural waters.

1.1.3 Limited research was conducted to determine the ranges of total and fecal coliform concentrations in natural waters worldwide. Results showed (Tables D-2 and D-3) that total coliform bacteria detected in natural waters ranged from non-detect to 500,000 cfu/100mL or 0 - 5×10^5 cfu/100mL. Based on the data collected, a background value of 1,000 – 10,000 (10^3 – 10^4) cfu/100mL was set as the target concentration of bacteria that must be introduced to the device continuously during non-sampling periods. Maintaining this concentration will create the constant challenge of bacteria that the device will encounter during field use, as well as give an indication of the ability of these organisms to "grow through" the device's bacteriological barrier over time. A more detailed description of the data is presented below.

1.2 A Review of the Bacterial Quality of Natural Waters

1.2.1 Natural water, as referenced here, is any body of water, standing or flowing, which is sufficient for a Warfighter to use as a water supply. A review of available data was performed to determine the bacterial quality of natural waters. The data reviewed primarily represented public water sources, such as lakes, large rivers, and reservoirs; recreational waters, and wadeable streams. The intent was to determine conditions under which a SWP must operate. The sources for this research include a number of scientific journals, the USEPA, the U.S. Geological Survey (USGS), the European Environment Agency, United Nations Environment Programme-Global Environment Monitoring System (GEMS) Water Programme, and the World Health Organization (WHO).

1.2.2 Source water standards exist in reference to recreational water safety and agricultural/fishery health impact. The USEPA, European Union Environment Programme, and the WHO recommend maximum bacteria concentrations as shown in Table D-1.

Table D-1. Recreational and Agricultural Water Standards (cfu/100mL).

Regulatory Agency	Total Coliform	Fecal Coliform	*E. Coli*	Fecal Strep	Enterococci	Note
USEPA		*200/400	126		33/35(Fresh /Marine)	*Previously accepted standards. Enterococci and *E. coli* are now used as indicator organisms
U.S. States Class A	100-2400	14-200	18-130		14-100	Classes are defined by each state. Generally, Class A reflects excellent quality for use such as shellfishing.
U.S. States Class C	2400	200-1000	142			Class C reflects Fair or minimal contact recreational use.
EU Imperative	10,000	2000				
EU Guideline		100		100		
WHO				40		This guideline represents the 95% level, below which less than 1 illness per 100 is probable.

1.2.3 Table D-2 shows the ranges for each organism found within the indicated body of water. As shown, concentrations vary considerably, indicating the effect that environmental or seasonal conditions have on water quality.

Table D-2. Data Gathered from Literature Review of Professional Journals Including Studies Conducted By USEPA, USGS, International Environmental Agencies, and Educational Institutions. (See references for select citations. Units are cfu/100mL unless otherwise noted.)

Location	Sample Date	Water Source	Total Coliform	E. Coli	Fecal Coliform
Kansas	1998-01	Flowing	105-43,900	360	1,400
Wyoming[3]	1990-99	Flowing			1-45,000
Massachusetts	2004	Flowing		1-2,900	
Sierra-Nevada	2004	Mixed	600-10,000		
Canyon Lake CA[4]	2002	Standing	8500[a]	1-15[a]	1,000[a]
Seoul, South Korea	2002	Running	228	0-10	
Brazil	2001-02	Standing	3200-40,000		
New Zealand					1,000
Germany[5]	2006	Standing		4.7-5,300	3-1,500
Epi Review	1950-Present	Standing	45-5,750	2-5,200	2-2,000
Benton, AK	1969-84	Mixed	0-35,000	1-2,100	0-50,000
Rio Grande[6]	2005	Running	27,000-95,000[a]		11,000-33,000[a]
Myponga Reservoir, Australia[7]	2003	Mixed		15-23,000	9-10,000
Rio San Juan, Mexico[8]	1995-96	Flowing	332,000[b]		111,000[b]
Epi study by Kay et al	1994	Standing			158 Strep
Grand River, Canada[9]	2002	Flowing	300-35,000	300-27,000	500-33,000

[a] Mean values
[b] Values may be per L vice per 100 mL, notation in journal unclear

1.2.4 Water treatment facilities in the U.S. use raw water quality information to determine appropriated treatment strategies for compliance with USEPA drinking water regulations. The Information Collection Rule of 1996 required the USEPA to compile data on the presence of certain microbes in addition to indicator organisms in drinking water sources. A summary of these is shown in Table D-3.

Table D-3. Summary Results of the Information Collection Rule Auxiliary Database 1, for the Presence of Selected Organisms in Raw Source Waters, 1997-1998. (Data represent results from U.S. states and territories. Summary values are for detections only. Non-detects are not included in summary calculations)

Organism	Min	Max	Mean	Median	Total Records	Non-detects
Giardia Cysts/100L	0.2	2,521	139	52	6,941	5,818
Crypto Cysts 100L	0.1	1,923	95	30	6,941	6,530
Total Coliform/100mL	1	500,000	1,037	4	9,636	4,287
Fecal Coliform/100mL	1	90,000	277	18	7,496	4,979
E. Coli/100mL	1	242,000	375	16	4,453	2,112
Virus/100L	0.1	1,974	16	2	4,207	3,332

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APPENDIX E
GROWTH CURVES FOR *E. COLI* AND *R. TERRIGENA*

1.1 Growth Curves

1.1.1 A 28-hour growth curve evaluation was performed in triplicate for *R. terrigena* and *E. coli*. The cultures were grown according to the procedures described in section 3.6.2.2. of this Protocol. Three flasks were prepared for each organism, and sampling was performed directly from each flask at each time point. The bacterial organisms were initially obtained from ATCC in a pelletized form and were rehydrated according to the manufacturer's instructions. A 1 mL aliquot of the resuspended culture was transferred to 10 mL of TSB and the suspension was incubated at 35 °C for 24 hr (Pass #1). The incubated cultures were then prepared for -70 °C storage by adding 10% glycerol to the growth suspension. A frozen vial of each culture was thawed and 0.1 mL of the thawed culture was transferred to 10 mL of TSB (Pass #2). The 10 mL suspension was incubated for 24 hr at 35 °C. Following 24-hour incubation, 5 mL of the challenge organism stock was amended to 500 mL TSB (Pass #3).

1.1.2 The growth suspension was placed on a rotary shaker at 150 RPM and incubated at 35 ºC for 16-18 hours. At each time point, 1.15 mL was removed from each flask for plating and spectrophotometric analysis. The absorbances of the cultures were recorded at 415 nm and 595 nm using a spectrophotometer (Biorad Model 680). 150-μL volumes were added per well, of a 96 well plate. 1 mL aliquots were used in a pour plating scheme to quantify the challenge organisms at each time point. Sterile buffered deionized water was used for the diluent and Standard Plate Count Agar was used for the growth medium for plating. The plates were incubated at 35 °C for 24 hours. The results of the growth study are presented below (Figures E-1 and E-2).

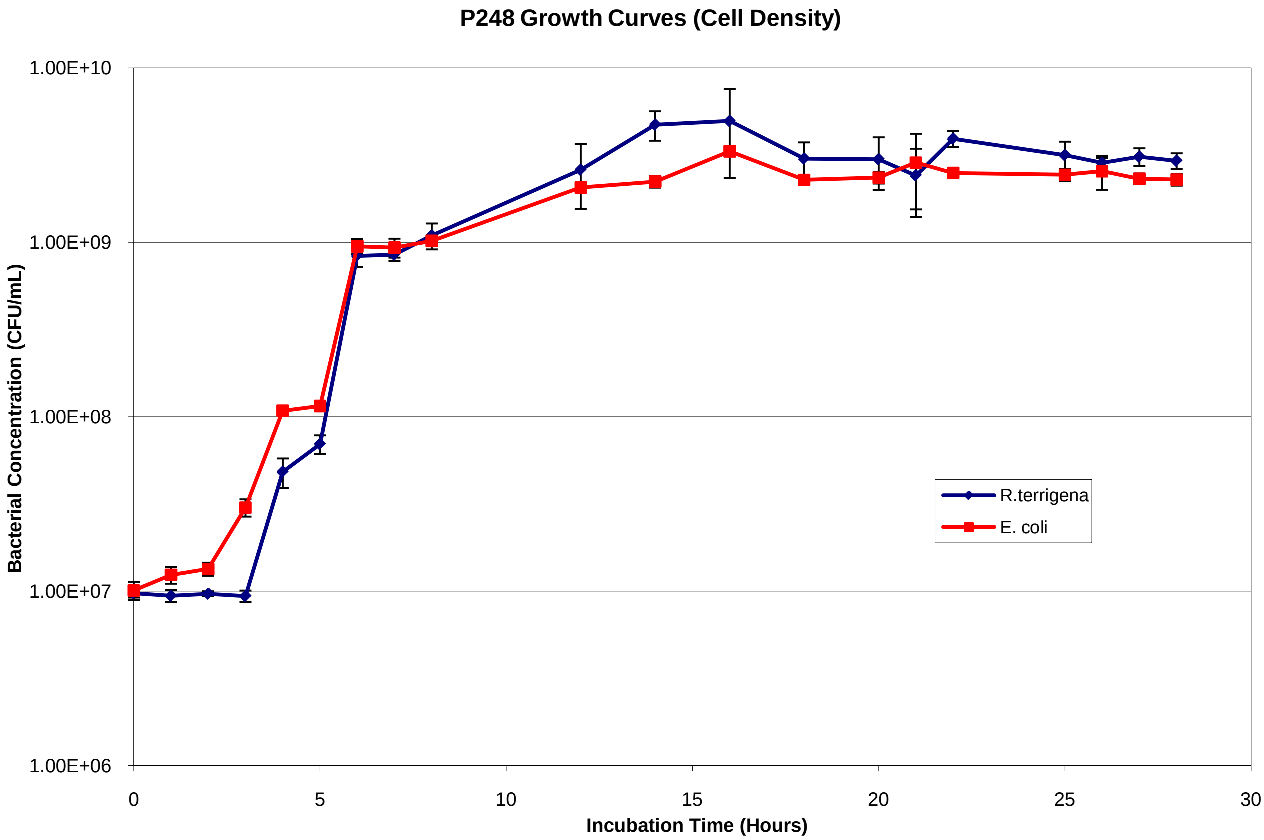

Figure D-1. Observed bacterial concentration for *R. terrigena* and *E. coli* over a 28-hour growth experiment. Individual points and error bars represent the means and standard deviations of three replicates per challenge organism. Logarithmic growth phase appears to occur between 4 and 9 hours post startup. Late log phase/early stationary phase is apparent at 16-18 hours post startup.

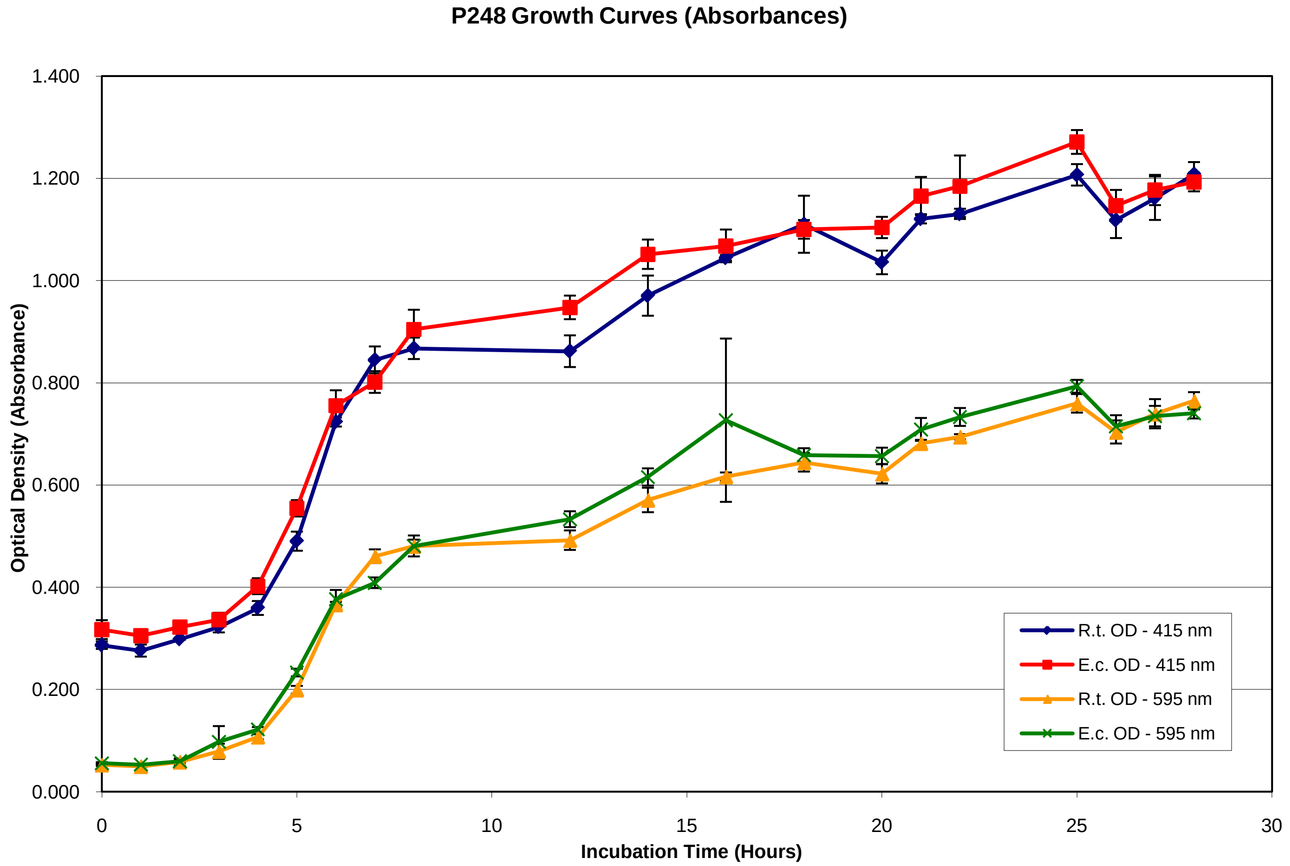

Figure D-2. Observed optical density values for *R. terrigena* and *E. coli* 28-hour growth study. Individual points and error bars represent the means and standard deviations of three replicates per challenge organism. Greater absorbances for each point were observed at 415 nm compared to 595 nm. Logarithmic growth phase appears to occur between 5 and 9 hours post startup. Late log phase/early stationary phase is apparent at 16-18 hours post startup.